

EXERCISE - 2

C Compiler in Virtual Machine

Aim:

To install the GCC C compiler in the Ubuntu virtual machine and execute a simple C program.

Algorithm:

1. Start the Ubuntu virtual machine.
2. Open the Terminal using **Ctrl + Alt + T**.
3. Update the Ubuntu package repository using:
`sudo apt update`
4. Install the build-essential package using:
`sudo apt install build-essential -y`
5. Verify the GCC installation using:
`gcc --version`
6. Create a C source file named **hello.c** using the Nano editor:
`nano hello.c`
7. Enter the following C program:

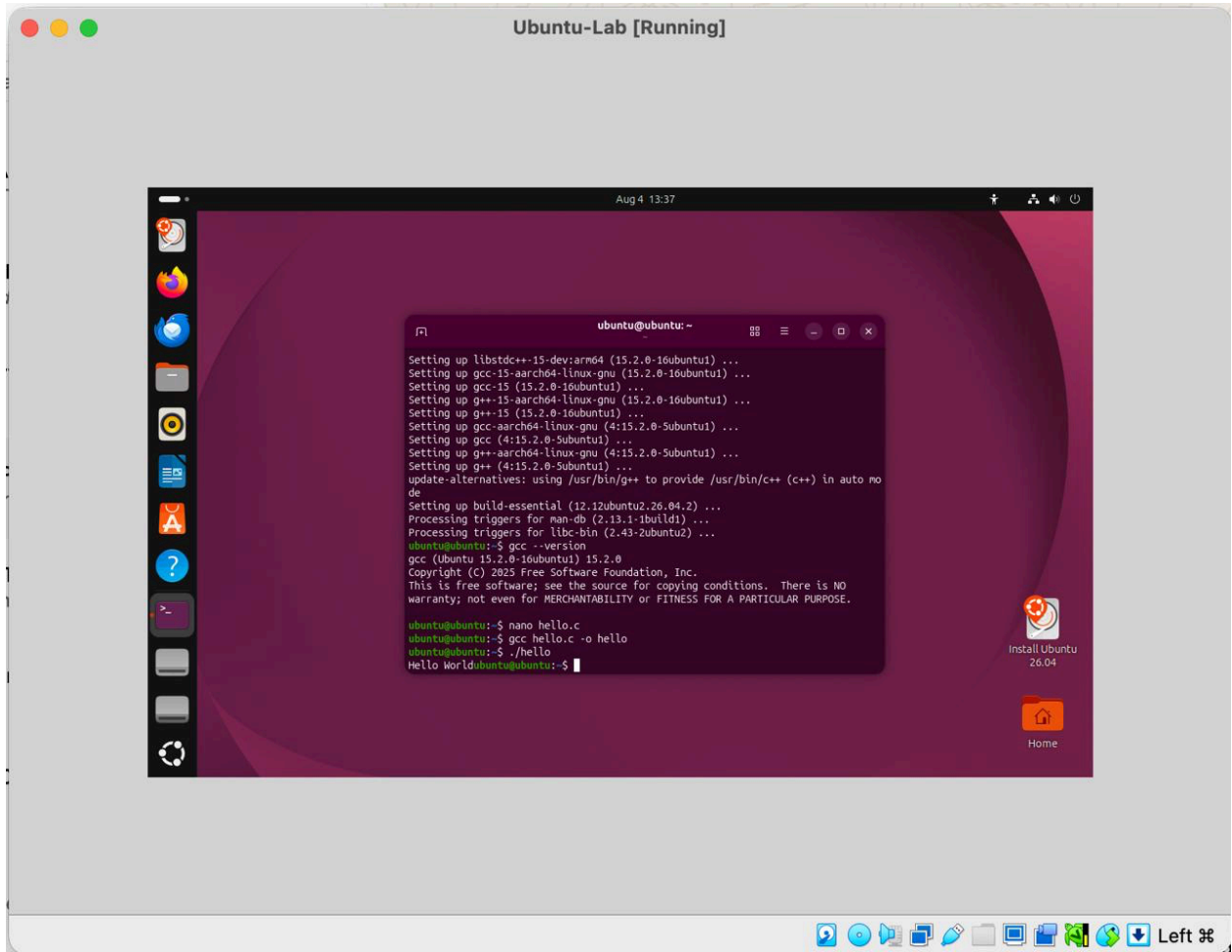
```
#include <stdio.h>

int main() {
    printf("Hello, Virtual Cloud World!\n");
    return 0;
}
```
8. Save and exit the file.
9. Compile the C program using:
`gcc hello.c -o hello`
10. Execute the compiled program using:

`./hello`

11. Verify the program output.

Output:



```
ubuntu@ubuntu:~$  
Setting up libstdc++15-dev:amd64 (15.2.0-16ubuntu1) ...  
Setting up gcc-15-aarch64-linux-gnu (15.2.0-16ubuntu1) ...  
Setting up gcc-15 (15.2.0-16ubuntu1) ...  
Setting up g++-15-aarch64-linux-gnu (15.2.0-16ubuntu1) ...  
Setting up g++-15 (15.2.0-16ubuntu1) ...  
Setting up gcc-aarch64-linux-gnu (4:15.2.0-Subuntui) ...  
Setting up gcc (4:15.2.0-Subuntui) ...  
Setting up g++-aarch64-linux-gnu (4:15.2.0-Subuntui) ...  
Setting up g++ (4:15.2.0-Subuntui) ...  
update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++ (c++) in auto mo  
de  
Setting up build-essential (12.12ubuntu2.26.04.2) ...  
Processing triggers for man-db (2.13.1-1build1) ...  
Processing triggers for libc-bin (2.43-2ubuntu2) ...  
ubuntu@ubuntu:~$ gcc --version  
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0  
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This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  
ubuntu@ubuntu:~$ nano hello.c  
ubuntu@ubuntu:~$ gcc hello.c -o hello  
ubuntu@ubuntu:~$ ./hello  
Hello Worldubuntu@ubuntu:~$
```

Result

Thus, the GCC C compiler was successfully installed in the Ubuntu virtual machine, and the C program was compiled and executed successfully.